

KARTHIKEYAN V R

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PROFESSIONAL SUMMARY

Computer Science graduate (CGPA 8.6) with hands-on full-stack and backend development experience across Java, Python, React.js, and FastAPI, built through a Software Developer Trainee role and applied engineering projects. Comfortable across the full SDLC — REST API development, authentication, database integration, testing, and debugging — with additional project experience in RAG pipelines and CNN-based computer vision.

TECHNICAL SKILLS

Languages: Java, Python, JavaScript, SQL

Frontend: React.js, Next.js, HTML, CSS, Bootstrap

Backend: FastAPI, REST APIs, JWT Authentication, Object-Oriented Programming

Databases: PostgreSQL, Supabase, SQL Querying

AI / ML: PyTorch, CNN, FAISS, Sentence-Transformers, Retrieval-Augmented Generation (RAG)

Cloud & Tools: AWS Cloud Foundations, AWS Cloud Operations, Git, CI/CD Fundamentals

PROFESSIONAL EXPERIENCE

Software Developer Trainee — Arunai Enterprises, Chennai, India

Mar 2026 – May 2026

- Developed React.js frontend modules for NexServe, a full-stack food service management application, covering authentication, food filtering, shopping cart, and checkout workflows.
- Built and contributed to REST API endpoints in FastAPI that supported core application workflows, and implemented JWT-based authentication.
- Integrated PostgreSQL/Supabase for persistent application data and added SMTP-based email notifications.
- Debugged frontend and backend issues and tested features end-to-end to verify functional correctness before release.
- Documented bugs, fixes, and testing outcomes, and supported deployment activities while collaborating with the team across the SDLC.

PROJECTS

AI-Powered Knowledge Base & RAG System — Best Project Award | Python, FastAPI, FAISS, Sentence-Transformers

- Built a retrieval-augmented generation (RAG) pipeline covering document ingestion, text processing, and embedding generation with Sentence-Transformers.
- Implemented vector similarity search with FAISS and exposed the pipeline through FastAPI endpoints; refined it through systematic debugging, earning the Best Project Award.

NexServe — Full-Stack Food Service Management System | React.js, FastAPI, Python, PostgreSQL/Supabase, REST APIs

- Built a full-stack web application enabling customer and vendor workflows, including food filtering, shopping cart, and checkout.
- Designed REST APIs with FastAPI, integrated JWT authentication, and connected the app to Supabase/PostgreSQL for data storage.

Advanced DeepVision — Skin Cancer Detection System | Python, PyTorch, CNN, FastAPI, Next.js

- Built an end-to-end system integrating a CNN-based image classification model with a Next.js frontend and FastAPI backend.
- Implemented image upload, preprocessing, and inference pipelines with REST APIs exposing classification and confidence scoring, plus Grad-CAM explainability visualization.

EDUCATION

B.E., Computer Science and Engineering — S.A. Engineering College, Chennai

2022 – 2026

CGPA: 8.6

CERTIFICATIONS

- CS50's Introduction to Artificial Intelligence with Python — Harvard CS50
- AWS Academy Graduate — Cloud Foundations and Cloud Operations
- NPTEL — Foundation of Cloud IoT Edge ML

ACHIEVEMENTS

- Best Project Award — AI-Powered Knowledge Base & RAG System
- 1st Prize — College Hackathon, for developing a full-stack web application
- 2nd Place — IEEE Day, Innovative Web Application